State Ownership and Deposit Safety

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Abstract

Do depositors see state-owned banks as safer? The isolated, unexpected failure of Banco Santos, a private bank, in November 2004 raised the perceived risk of uninsured deposits in Brazil. Time deposits at state-owned banks grew 6.6 percentage points per month faster than at private banks, a gap of 56 percent after seven months. Excluding the largest banks leaves the estimate unchanged, separating ownership from too-big-to-fail size. Demand and savings deposits, whose clients were almost all fully insured, show no movement.

JEL classification: G01; G21; G28
Keywords: State-owned banks; Uninsured deposits; Government ownership; Bank runs; Flight to safety; Too big to fail

1 Introduction

When a bank fails, where does the money go? Depositors punish weak banks and reward banks they see as safe (Martinez Peria and Schmukler, 2001; Iyer and Puri, 2012). The best-documented safety attribute is size: in systemic episodes, deposits flee to the largest banks, which depositors treat as too big to fail (Oliveira et al., 2015). I ask whether depositors also respond to *government ownership*: when deposits become risky, do they flow to state-owned banks?

Separating ownership from size is hard. In Brazil, as in other emerging markets, the largest banks are both private (Itaú, Bradesco) and state-owned (Banco do Brasil, Caixa Econômica Federal): a flight toward state banks could simply be a flight toward big balance sheets. But Brazil also has small state-owned banks, so ownership can be observed separately from size.

Banco Santos, a mid-sized private São Paulo wholesale bank, held 0.4% of system deposits (demand plus savings plus time deposits). Its funding was almost exclusively (97%) large time deposits of companies and pension funds, mostly far above the insurance limit of the time (R$20,000).[1] Its collapse in November 2004 followed fraud, and no other bank failed for a year: an isolated shock to perceived deposit risk, its only role in the analysis (Section 3).

In a difference-in-differences design on every bank in Brazil, time deposits at state-owned banks grew 6.6 percentage points per month faster than at private banks over the event month and the six months after ($p < 0.01$), a compounded gap of 56 percent for the typical state-owned bank. The estimate barely moves (6.9 points) when Banco do Brasil, Caixa, and all other large banks are excluded, so that only small banks remain: there, the flight was toward ownership.

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[1] Among documented creditors, pension funds held R$5–38 million each at the intervention (Agência Câmara de Notícias, 2004).

Wild-bootstrap and randomization inference agree; no single bank drives the result. Demand and savings deposits, whose holders were nearly all fully covered, show nothing: the reallocation happened only where uninsured money sits.

The literature on government ownership of banks is mostly about assets: politically allocated credit (La Porta et al., 2002; Sapienza, 2004; Dinç, 2005), less procyclical lending (Micco and Panizza, 2006; Brei and Schclarek, 2013), stabilized Brazilian credit in 2008–09 (Coleman and Feler, 2015). On liabilities, evidence from India shows deposits moving toward public-sector banks in stress episodes (Eichengreen and Gupta, 2013; Acharya and Kulkarni, 2019; Acharya et al., 2022). Relative to that work, the shock here is idiosyncratic rather than systemic, and the design separates ownership from size, impossible where public banks are also the largest, as in India or Brazil in 2008 (Oliveira et al., 2015). The funding advantage I document arises from depositor choices (Calomiris and Jaremski, 2016): no explicit guarantee covers these deposits at either type of bank.

2 The episode, data, and sample

The episode. The Central Bank intervened in Banco Santos on 12 November 2004 and liquidated it the following year. In October 2004 it held R$1.74 billion in deposits, 97% of it in time deposits, and essentially no demand or savings business. FGC (*Fundo Garantidor de Créditos*) insurance covered R$20,000 per depositor per bank, ample for retail savers, far below Santos's institutional claims.[2] The intervention uncovered sham loans to shell companies and off-book liabilities; the controller was later sentenced to 21 years for financial crimes. Brazil that year had robust growth, falling inflation, and no other collapse.

Data. ESTBAN, the Central Bank's monthly branch-level statistics (also used by Barros Jr. et al. 2025), aggregated to the bank level, measures time deposits (*depósitos a prazo*), the largely uninsured margin; demand deposits (*conta corrente*); and savings (*caderneta de poupança*). Banks are state-owned (federal or state-government) or private, classified by tax identifier with time-varying windows over the 1997–2005 privatizations (Beck et al., 2005).

Sample. The sample has 84 banks, 13 state-owned; Appendix Tables A.6 and A.1 trace the exclusions and describe the groups. *Small* banks each hold under 2% of system deposits seven months before the event; together, the ten small state-owned banks, led by Banrisul and Banco do Nordeste, hold 3.6% in the month before the event. Small private banks are the run-exposed segment: the median one funds 97% of its deposits with time deposits, the product where nearly all above-ceiling money sits (footnote 2), a funding profile like Santos's.

3 Empirical strategy

The treatment is not the failure of one bank but the change in perceptions it caused. In the spirit of Oliveira et al. (2015), I need a shock sharp enough to shift behavior abruptly. The sudden discovery of fraud at a bank that had reported sound accounts raised, at once, the perceived risk of uninsured claims on private banks. The regressions measure the differential response by ownership. I do not use 2008, when global turmoil moved far more than perceptions of bank safety.

[2]Time deposits were insured up to the same ceiling; they are simply where uninsured money sat (census coverage shares under both ceilings: Appendix Table A.1) (Banco Central do Brasil, 2006a,b).

For bank b and month t from six months before to six months after the intervention, I estimate

$$\Delta \log Dep_{bt} \;=\; \alpha_b + \gamma_t + \beta\, State_b \times Post_t + \varepsilon_{bt}, \tag{1}$$

where the outcome is monthly log growth, $State_b$ indicates government ownership, and $Post_t$ turns on in the event month. Growth rather than the level keeps bank-specific trends out of the estimate; it is winsorized at 1st/99th. Standard errors cluster by bank, where treatment is assigned. With only thirteen treated banks (ten in the small sample), Table 1 also reports a wild cluster bootstrap and randomization inference that reassigns which banks are labeled state-owned. Each margin is estimated on all banks and the small sample.

To trace dynamics and pre-trends, I estimate the event-study version

$$\Delta \log Dep_{bt} \;=\; \alpha_b + \gamma_t + \sum_{\tau \neq -1} \beta_\tau\, State_b \times \mathbf{1}[t - t^* = \tau] + \varepsilon_{bt}, \tag{2}$$

over $\tau = -6$ to $+12$, where t^* is the event month; the previous month is omitted. A single month is a noisy reference, so I re-center the coefficients, denoted $\tilde{\beta}_\tau$, so that pre-event months average zero, a pure normalization. Figure 1 plots the running sum $C_\tau = \sum_{s=0}^{\tau} \tilde{\beta}_s$, which recovers the implied log-level event-study path from the first-differenced specification (Schmidheiny and Siegloch, 2023): the gap against a counterfactual continuing the pre-event average differential, with exact clustered standard errors. The quarterly version and other margins are in Appendix Figures A.2–A.3.

4 Results

Table 1: Deposit flows to state-owned banks around the failure

	(1) Time deposits	(2) Demand deposits	(3) Savings
Panel A. All banks			
State-owned × Post	0.064***	−0.011	−0.003
	(0.012)	(0.027)	(0.005)
Wild bootstrap p	<0.001	0.696	0.613
Randomization p	<0.001	0.735	0.629
Observations	1,038	1,025	364
Banks (state-owned)	84 (13)	83 (13)	32 (12)
Panel B. Small banks only			
State-owned × Post	0.067***	−0.012	−0.004
	(0.014)	(0.031)	(0.008)
Wild bootstrap p	<0.001	0.696	0.649
Randomization p	<0.001	0.733	0.669
Observations	908	895	247
Banks (state-owned)	74 (10)	73 (10)	23 (9)
Bank FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Notes: β from Eq. (1), $\tau = -6$ to $+6$; monthly log growth in log units, converted to percentage points and percent in the text; winsorized 1st/99th. Panel B: banks below 2% of system deposits at $\tau = -7$. Bank and month fixed effects; standard errors clustered by bank. Wild bootstrap: restricted, Webb weights, 9,999 draws. Randomization: 4,999 reassignments of the state-owned label across banks, studentized. Bank counts differ because growth requires a positive book. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 1 has the main result: 6.6 percentage points per month in the full sample ($p < 0.01$), compounding to 56 percent for the typical state-owned bank over seven months; 6.9 points and 60 percent in the small sample. Excluding the ten large banks changes almost nothing; the bootstrap and randomization p-values confirm both. Small state banks gained like the giants. Demand and savings differentials are −1.1 and −0.3 percentage points per month, with similar zeros in Panel B, far from significance: the run operated only where uninsured money sits.

Appendix Table A.2 puts the magnitude in perspective. The regression weights banks equally, and for the typical small bank the gap is 60 percent. The money moved less, because the group totals of Table A.2 weight banks by size: to month +6 the ten small state banks gained R$1.1 billion of time deposits (+12.6%) while small private banks lost R$3.9 billion (−10.2%), an aggregate differential near 25 percent; nearly half is the private-side decline, not Santos's R$1.7 billion book recycled. It ran alongside the flight to size: large private banks gained R$24.9 billion. The ownership margin is identified within the small segment and survives a size × Post control (Table A.4).

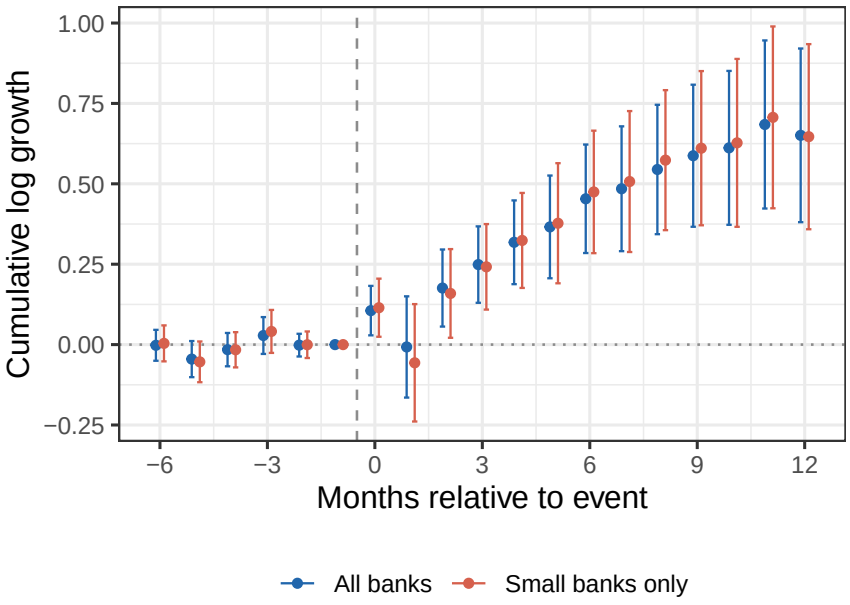


Figure 1: Cumulative event study, time deposits

Notes: Eq. (1) with event-time indicators ($\tau = -1$ omitted) over $\tau = -6$ to $+12$; coefficients re-centered so pre-event months average zero; each point is their running sum. Cluster-robust standard errors; bars are 95% confidence intervals.

Figure 1 shows flat pre-event coefficients: state and private banks were on parallel trends. The gap opens in the failure month and is significant from the first quarter. Differential growth runs at 5–7 percentage points per month for three quarters, then fades (Appendix Figure A.3); demand and savings stay flat (Figure A.2). The cumulative line drifts up through +12 because a running sum keeps early gains and adds the smallest banks' later moves; the aggregate differential peaks near +6 and roughly halves by +12 (Table A.2). The failure produced a one-time reallocation, persistent but partially mean-reverting, not a permanent change in growth rates.

The difference-in-differences coefficient for time deposits (Table 1, Panel B) is stable, staying between 0.056 and 0.071 across: dropping each state bank in turn; size cutoffs of 1% or 5%; no or tighter winsorization; a control group matched three-to-one on size; a size × Post control; and a full size × State × Post interaction, with the size terms never significant (Appendix Tables A.3–A.5). Adding the 24 foreign-controlled banks leaves the state coefficient unchanged, and these subsidiaries of global groups gained even faster (13 percentage points per month); the losers were the domestic private wholesale banks.

5 Depositor choice or administrative reallocation?

Santos's victims included municipal pension funds (Fala São Sebastião, 2022), and the scandal created pressure to keep public money at official banks. Did depositors move this money, or did administrators? ESTBAN reports time deposits as one line, so holders cannot be separated, but administrative placement fits the facts poorly. Public cash balances were already required to sit at official banks (Constitution, art. 164), and the post-scandal overhaul of pension-fund investment rules came only in October 2007 (CMN Resolution 3,506). The differential opens

in the failure month and spans state banks in every region, most with no link to the São Paulo entities that lost money at Santos. And demand deposits, where administered balances demonstrably sit, show no inflow. The reallocation looks depositor-driven.

6 Conclusion

An isolated failure that raised the perceived risk of uninsured deposits moved that money toward state-owned banks, toward tiny state banks as much as toward the federal giants, and not at all where depositors were fully covered. Holding size fixed, depositors treated state-owned banks as the safer place for money the guarantee did not cover. Why state banks are seen as safer, the data do not say. The asset-side stabilization by government banks (Micco and Panizza, 2006; Coleman and Feler, 2015) has a liability-side twin: state banks gain funding as private banks lose it, draining small private banks when they are most fragile (Egan et al., 2017).

Declarations

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Data availability: ESTBAN and the FGC census of guaranteed credits are publicly available from the Central Bank of Brazil; replication code is available from the author on request.
Declaration of generative AI in the writing process: the author wrote the code that produces the results and wrote the manuscript, and used Claude (Anthropic) to check both; after using it, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.
Supplementary material: an online appendix contains the additional tables and figures referenced in the text.

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Online Appendix

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Table A.1: Ownership groups in October 2004 (month before Banco Santos failed)

		Total deposits		Time deposits		Savings	
Group	Banks	R$ bn	%	R$ bn	%	R$ bn	%
Large private	7	187.3	39.7	96.2	39.2	56.9	37.5
Small private	64	46.0	9.8	38.5	15.7	3.1	2.0
Large state-owned	3	203.3	43.1	87.1	35.5	84.6	55.8
Small state-owned	10	17.0	3.6	8.9	3.6	5.4	3.6
Banco Santos (failed, excluded)	1	1.74	0.37	1.70	0.69	0.00	0.00

Notes: Banks in the estimation sample, grouped by ownership and size, one month before the intervention. *Small* banks hold less than 2% of system total deposits seven months before the event ($\tau = -7$); small state-owned and small private banks form the Panel B sample of Table 1 in the paper. The large state-owned group comprises Banco do Brasil, Caixa Econômica Federal, and Banco Nossa Caixa; the small state-owned group comprises Banrisul, Banco do Nordeste, Banestes, BRB, Banco da Amazônia, BESC, BEC, Banese, Banpará, and BEP. For total, time, and savings deposits, the table reports group holdings (R$ bn) and the share (%) of the system-wide aggregate. The last row shows Banco Santos, excluded from estimation. FGC coverage (semiannual census of guaranteed credits, December 2006 vintage): above-ceiling balances were 92.2% of time-deposit value under the R$60,000 census-date ceiling (97.0% under the R$20,000 ceiling in force at the event), against 58.6% (70.1%) of demand-deposit and 30.7% (56.0%) of savings value; over 97% of demand and savings clients were fully covered under either ceiling, against 76.7–88.9% of time-deposit clients (Banco Central do Brasil, 2006a,b).

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Table A.2: Aggregate time-deposit flows by ownership group

		Time deposits (R$ bn)			$\Delta \log \times 100$	
Group	Banks	$\tau = -1$	$\tau = +6$	Δ R$ bn	to +6	to +12
Large private	7	96.2	121.1	+24.9	+23.1	+31.3
Small private	62	38.0	34.1	−3.9	−10.8	+5.4
Large state-owned	3	87.1	101.0	+13.9	+14.8	+27.8
Small state-owned	10	8.9	10.0	+1.1	+11.9	+17.9
All state-owned	13	96.0	111.0	+15.0	+14.5	+26.9
All private	69	134.1	155.2	+21.0	+14.6	+24.6
System	82	230.2	266.2	+36.1	+14.6	+25.6

Notes: Aggregate time deposits, in reais, of the banks in the estimation sample with a complete time-deposit series over $\tau = -6$ to $+12$ (82 of 84 banks). $\Delta \log \times 100$ is the log change of the group aggregate from $\tau = -1$ to the indicated month. The small-state versus small-private aggregate differential is about 25 percent at $\tau = +6$ and 13 percent at $\tau = +12$: it peaks around month six and then roughly halves as small private books recover. Group aggregates weight banks by size, so they differ from the regression estimates, which weight banks equally; the regression measures the growth of the typical bank, the aggregate the growth of the money.

Table A.3: Leave-one-out: dropping each small state-owned bank

Bank excluded	Time deposits (R$ mn)	Coefficient	(s.e.)	p
BCO DO ESTADO DO RS S.A.	3,363	0.0682***	(0.0145)	0.0000
BCO DO NORDESTE DO BRASIL S.A.	2,039	0.0690***	(0.0143)	0.0000
BCO BANESTES S.A.	962	0.0710***	(0.0138)	0.0000
BRB - BCO DE BRASILIA S.A.	707	0.0679***	(0.0145)	0.0000
BCO DA AMAZONIA S.A.	528	0.0636***	(0.0141)	0.0000
BCO DO EST. DE SC S.A.	398	0.0653***	(0.0145)	0.0000
BCO DO EST. DO PA S.A.	352	0.0682***	(0.0145)	0.0000
BCO DO EST. DE SE S.A.	350	0.0649***	(0.0144)	0.0000
BCO BEC S.A.	187	0.0595***	(0.0123)	0.0000
BCO DO EST. DO PI S.A.	37	0.0699***	(0.0141)	0.0000
Baseline (none excluded)		0.0668***	(0.0138)	0.0000

Notes: Panel B time-deposit specification of Table 1 in the paper, re-estimated dropping one small state-owned bank at a time. Time deposits (R$ mn) as of October 2004. Standard errors clustered by bank. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A.4: Robustness of the small-sample time-deposit estimate

Specification	Coefficient	(s.e.)	p	Banks
Baseline (2% cutoff, winsorized 1/99)	0.0668^{***}	(0.0138)	0.0000	74
Size cutoff 1%	0.0708^{***}	(0.0146)	0.0000	71
Size cutoff 5%	0.0635^{***}	(0.0129)	0.0000	79
No winsorization	0.0557^{***}	(0.0173)	0.0019	74
Winsorized 2.5/97.5	0.0665^{***}	(0.0135)	0.0000	74
Size-matched controls (3:1)	0.0657^{***}	(0.0163)	0.0003	40
Controlling for size × Post	0.0639^{***}	(0.0149)	0.0001	74
Adding size × State × Post	0.0680^{***}	(0.0188)	0.0005	74

Notes: Variants of the Panel B time-deposit specification of Table 1 in the paper. Size cutoffs redefine the small sample at 1% or 5% of system deposits at $\tau = -7$. Winsorization variants apply no winsorization or 2.5/97.5 winsorization to monthly log growth. The matched specification keeps the ten small state banks and, for each, the three private banks closest in log total deposits at $\tau = -7$, chosen without replacement. The last two rows add $size_b \times Post_t$ ($size_b$ = log total deposits at $\tau = -7$, centered) and, in the final row, also $State_b \times size_b \times Post_t$; the reported coefficient is State × Post, at mean size in the final row. The size terms are never significant ($p \geq 0.57$). Standard errors clustered by bank. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A.5: Foreign-controlled banks as a third ownership group

	(1) Time deposits	(2) Demand deposits	(3) Savings
Panel A. All banks			
State-owned × Post	0.064***	−0.010	−0.003
	(0.012)	(0.027)	(0.005)
Foreign × Post	0.123***	−0.026	0.016
	(0.037)	(0.047)	(0.009)
Observations	1,315	1,281	416
Banks	108	104	36
Panel B. Small banks only			
State-owned × Post	0.067***	−0.012	−0.004
	(0.014)	(0.031)	(0.007)
Foreign × Post	0.123***	−0.028	0.017
	(0.037)	(0.049)	(0.011)
Observations	1,185	1,151	299
Banks	98	94	27
Bank FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Notes: Eq. (1) of the paper augmented with $Foreign_b \times Post_t$; the omitted group is domestic private banks. The sample adds the 24 foreign-controlled institutions otherwise excluded, mostly wholesale subsidiaries of global banking groups; all pass the same screens as the baseline sample. Bank and month fixed effects; standard errors clustered by bank. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A.6: Sample construction

	Banks excluded	Banks remaining
Institutions reporting in the window	–	141
Banco Santos and previously failed banks	1	140
Ownership class changes inside the window	0	140
Terminal wind-down (final six months of operation)	3	137
Mean time deposits below R$1 million	26	111
Foreign-controlled institutions	24	87
Unclassified institutions	3	84
Estimation sample (Panel A)	–	84
Small sample (Panel B)	10	74

Notes: Sequential exclusions over the estimation window ($\tau = -6$ to $+6$). 141 institutions report deposits at some point in the window (138 in October 2004 itself). "Previously failed banks" removes Banco Santos; banks that failed in earlier episodes contribute no window observations to begin with. The small-bank exclusion in the last row drops the banks at or above 2% of system deposits at $\tau = -7$: Banco do Brasil, Caixa Econômica Federal, Banco Nossa Caixa, and seven large private banks.

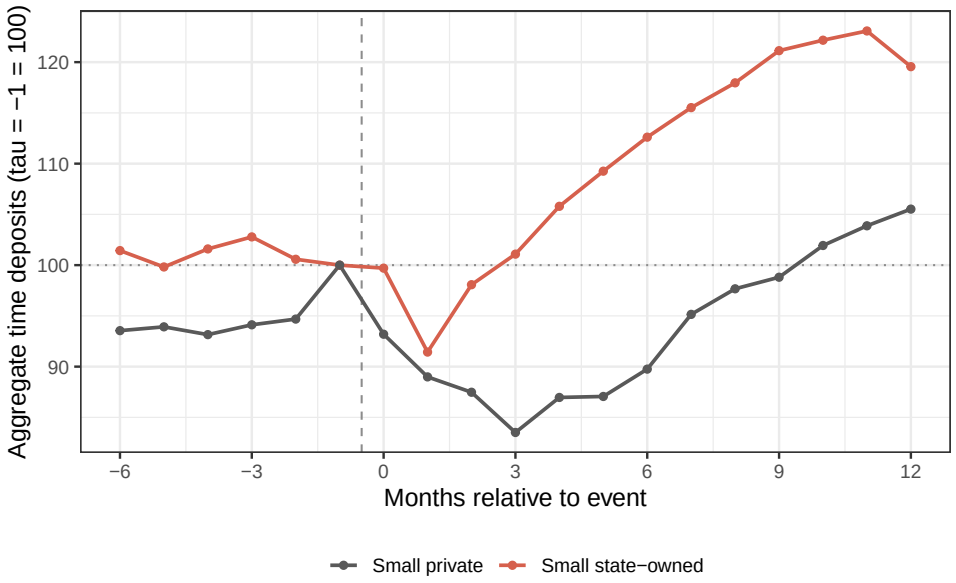
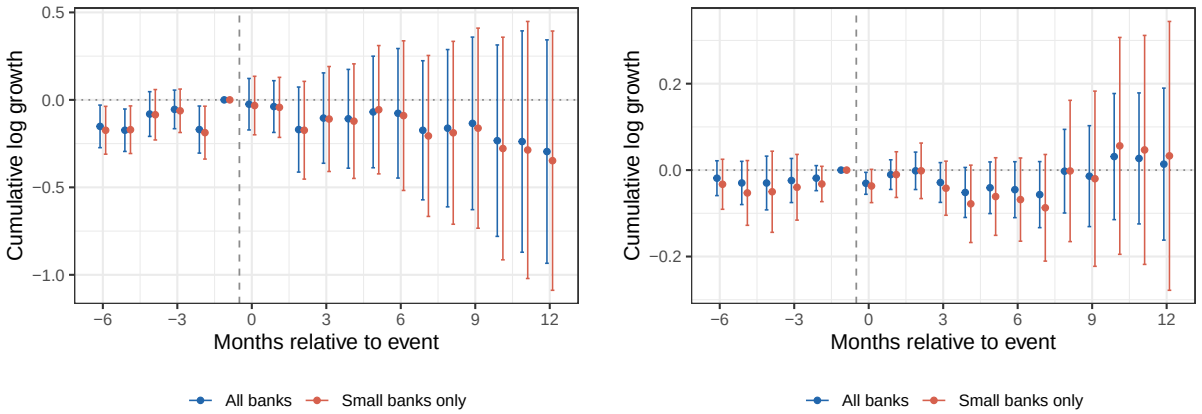


Figure A.1: Aggregate time deposits, small state-owned vs small private banks

Notes: Aggregate time deposits of the small state-owned and small private groups (banks with a complete series over $\tau = -6$ to $+12$), indexed to 100 in October 2004 ($\tau = -1$). The vertical dashed line marks the intervention in Banco Santos.

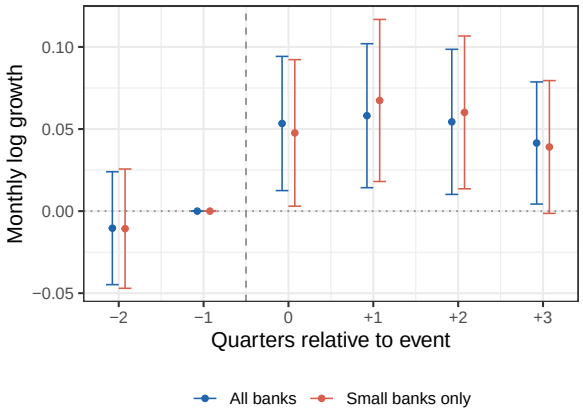


(a) Demand deposits

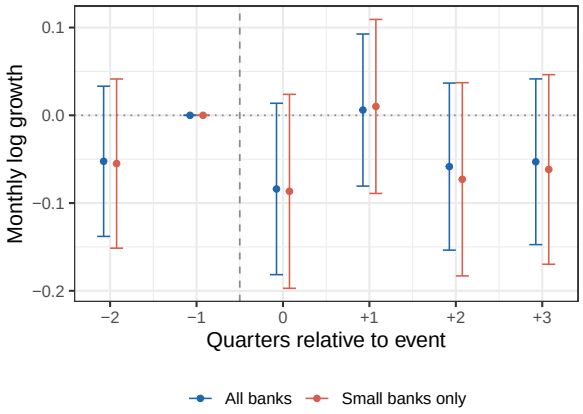
(b) Savings (*poupança*)

Figure A.2: Cumulative event studies, insured margins

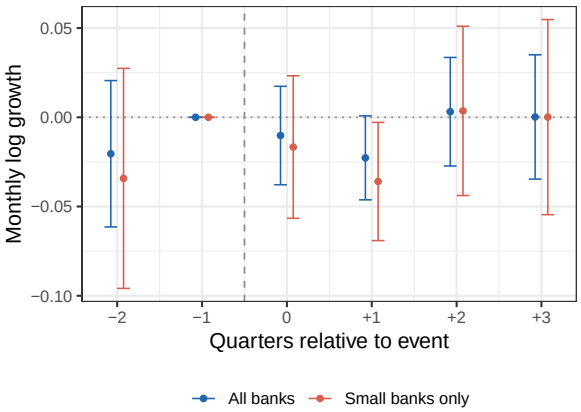
Notes: As Figure 1 in the paper, for demand and savings deposits. Coefficients re-centered so pre-event months average zero; each point is the running sum; cluster-robust (by bank) standard errors; bars are 95% confidence intervals.



(a) Time deposits



(b) Demand deposits



(c) Savings (*poupança*)

Figure A.3: Quarterly event-study estimates

Notes: Eq. (2) of the paper with event time binned into three-month quarters over $\tau = -6$ to $+11$ (0 = months 0 to +2, etc.), the quarter before the event as the omitted reference; each coefficient is the average differential monthly growth in that quarter. Bank and month fixed effects; standard errors clustered by bank; bars are 95% confidence intervals. Samples as in Table 1 in the paper.

References

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